FlowBridge

Setting up FlowBridge on a Synology

Step by step, from nothing to live readings from your EcoFlow power station. No prior knowledge, no command line — everything runs through the NAS interface.

DSM 7 · CONTAINER MANAGER | 5 TO 10 MINUTES | TWO ROUTES – ONE IS ENOUGH

SO WE MEAN THE SAME THINGS

Three words that keep coming up

This is all the jargon you need for this guide. If the terms are familiar, just skip the box.

- **Image**
  A finished software package to download — comparable to an installer, except everything the program needs to run is already inside. You do not install anything, you only download it.
- **Container**
  Such an image while it is running. It lives in its own little box on the NAS and never touches the rest of your system. Delete the container and it is gone without a trace.
- **Broker**
  The exchange for MQTT — the software where all the devices in your house drop off and pick up their readings. Usually Mosquitto, often part of Home Assistant or EisBär. FlowBridge delivers there.

PREPARATION

Three things to fetch beforehand

Get these now. Going looking for them halfway through the setup is the surest way to lose the thread.

- **An access key and secret key from EcoFlow**

  Two long strings with which FlowBridge identifies itself to EcoFlow. You get them free of charge on the EcoFlow IoT Open Platform (https://developer-eu.ecoflow.com): register, wait briefly for activation, then both keys are shown in the portal.

  Important: sign in there with the same account your power station is registered to in the EcoFlow app. With the keys of a different account FlowBridge will not find your device.

- **The address of your MQTT broker**

  Its IP address on your own network, usually something like `192.168.1.50`, plus the port number — almost always `1883`. If your broker requires a user name and password, have those ready too.

- **Container Manager on the NAS**

  Check in **Package Center** whether it is installed. If not: search, install, done. That takes two minutes and needs no restart.

HERE WE GO

Two steps for everyone, then one decision

You need the next two steps in any case. After that there are two routes to get FlowBridge onto the NAS — **one is enough**, both end at the same running container.

IF THE LABELS LOOK DIFFERENT

This guide uses the English DSM wording. Depending on the DSM version the menu entries can be named slightly differently. What matters are the things themselves, not the exact word.

1 Create two folders

Open **File Station** — the file manager of the NAS. There is a shared folder named `docker`. If there is not, create it.

Inside it, build this structure:

```
docker › flowbridge › data
```

So one folder `flowbridge`, and **inside that** another one named `data`. Both names exactly like that, lower case.

WHY TWO FOLDERS

In `flowbridge` the NAS will put its own project file. In `data` your credentials and settings end up. Keeping them apart pays off later: you can delete the project and create it again without losing your setup.

YOU DO NOT NEED TO WORRY ABOUT PERMISSIONS

Inside the container FlowBridge deliberately does **not** run as system administrator — that is the one real safeguard a container brings. So that the data folder is writable regardless, FlowBridge prepares it itself at startup.

You therefore need neither SSH nor hand-assigned permissions. Should it get stuck anyway, FlowBridge says in the browser, in plain words, what to do — instead of quietly restarting in circles.

2 Open Container Manager

Start **Container Manager** from the main menu of the NAS. In the sidebar you will find several entries — **Overview**, **Container**, **Image**, **Registry**, **Project**.

Two of them lead to the goal: **Project** (route A) and **Registry** (route B). Which one to take is explained right below.

TIME TO DECIDE

Route A or route B?

Both fetch the same image from the same place. The only difference is **how** the settings get in: as one block of text to paste — or clicked through several dialog pages.

	ROUTE A – WITH TEXT	ROUTE B – VIA THE REGISTRY
Time	about 5 minutes	about 10 minutes
What you do	copy one block, change one word	search for the image, then fill in fields across several pages
Where it goes wrong	the indentation when pasting	port, folder path and variable names — each by hand
Log size capped	**yes**, it is in the text	**no**, the wizard does not offer it
Health check	**yes**	no

WHY I RECOMMEND ROUTE A

Not because of the five minutes. The block of text carries two settings the wizard does not offer at all: a **cap on the log files** (3 × 10 MB) and a **health check**, by which the NAS notices when FlowBridge is running but no longer answering.

Without the cap the container log keeps growing indefinitely — on a NAS that runs for years, that is not a theoretical problem.

Route B is described anyway: not everyone likes YAML blocks, and whoever wants to see what the NAS actually configures sees it there field by field.

ROUTE A – RECOMMENDED

With ready-made text

You create a **project** and paste in a finished description. It holds everything at once: which image, which port, which folder, which password.

1 Create the project

In Container Manager click **Project** in the sidebar, then **Create** at the top.

```
Container Manager › Project › Create
```

A form opens. Fill it in like this:

Create project

Project name	`flowbridge` Lower case, no spaces.
Path	Click **Set Path** and choose `docker/flowbridge` The outer folder — **not** `data`.
Source	„Create docker-compose.yml“ Usually preselected. Do not choose „upload“ — you do not have a file, you have text.
Large text field	Leave empty for now — the content comes in the next step

IF THE LABELS DIFFER

Depending on the DSM version the fields are named slightly differently. What matters are the three things: **name**, **folder** and choosing to **enter the text yourself** instead of uploading a file.

2 Paste this text

This is the description the NAS starts FlowBridge from. Select all of it, copy it and paste it into the large text field. You will change the **orange** line in a moment.

DOCKER-COMPOSE.YML — COPY IN FULL

```
services:
  flowbridge:
    image: cheetahlab/flowbridge:latest
    container_name: flowbridge
    restart: unless-stopped

    ports:
      # Left is the port on the NAS, right the one in the container.
      # Only change the left number, if 8081 is already taken.
      - "8081:8080"

    environment:
      FLOWBRIDGE_CONFIG: /config/config.yaml
      TZ: Europe/Berlin
      FLOWBRIDGE_PASSWORD: "put-a-good-password-here"

    volumes:
      # Left your folder on the NAS, right the place in the container.
      - /volume1/docker/flowbridge/data:/config

    healthcheck:
      test:
        - CMD
        - python
        - -c
        - "import urllib.request;
urllib.request.urlopen('http://127.0.0.1:8080/api/auth/state', timeout=5)"
      interval: 30s
      timeout: 10s
      start_period: 20s
      retries: 3

    logging:
      driver: json-file
      options: { max-size: "10m", max-file: "3" }

    stop_grace_period: 20s
```

Now set the password

Replace `put-a-good-password-here` with one of your own. **Keep the quotation marks.** So

```
FLOWBRIDGE_PASSWORD: "put-a-good-password-here"
```

becomes, for example

```
FLOWBRIDGE_PASSWORD: "Sunshine-2026!"
```

WHY ALREADY NOW

FlowBridge can not only read your power station but also control it — start charging, switch outputs, change charging power. Without a password that would be open to anyone on your network.

Entering it here means the protection is active from the very first start. There is then no window in which FlowBridge sits unprotected on the network.

MIND THE INDENTATION

When pasting, please do not change the spaces at the start of the lines — this file format takes indentation literally. If the NAS complains about the description, almost always a line has slipped. Then simply copy it again cleanly.

3 Through the wizard and start

From here the NAS leads you through a few pages. In order:

PAGE	WHAT YOU DO
Web Portal / Web Station	Nothing. You do not need this — just **Next**.
Summary	Glance over whether path and name are right, then **Done**.
Log window	Runs by itself. Watch it or get a coffee.

Now the NAS downloads the image, a good 200 MB. Lines rush past in the log window — that is normal and need not concern you. At the end it says, in effect, that the container has been created and started.

Once the project shows **Running** in the overview, the hardest part is behind you.

NO ACCOUNT NEEDED

You do not have to sign in anywhere or unlock anything. FlowBridge is publicly available and is simply downloaded.

ROUTE B — THE MANUAL ONE

Via the registry

No text, but more clicking: you search for the image in the registry, download it and configure the container through the wizard.

ALREADY DONE ROUTE A?

Then skip this section and continue at **Setting up FlowBridge**. Going both routes achieves nothing — the port would afterwards be taken twice.

1 Download the image

In Container Manager click **Registry** in the sidebar. There is a search field at the top — enter:

```
cheetahlab/flowbridge
```

The result appears after a few seconds. Select it, then click **Download**. It asks you for the tag.

WHICH TAG?

`latest` is at the top of the list, below it version numbers such as `2026.08.16-46`. For the first time `latest` is fine — that is always the newest one.

Updating later is a different matter: then the number is what counts, on both routes. Why, is explained below under **Updating to a new version**.

The progress is shown under **Image**. A good 200 MB; once the entry sits there without a progress bar, it is finished.

2 Configure the container

Click **Image** in the sidebar, select the downloaded one, then **Run** at the top. Now come the pages where route A simply had a block of text.

Create container

Container name	`flowbridge` Lower case, no spaces.
Auto-restart	enable it Otherwise FlowBridge will not start by itself after the NAS reboots.

Then click **Advanced Settings** — without those, port, folder and password are missing, and FlowBridge would be neither reachable nor able to remember anything.

Advanced Settings

Port Settings	Local port `8081` → container port `8080` The field often sits on „Auto“ — that yields a different number at every start. Enter 8081 explicitly.
Volume / Folder	`docker/flowbridge/data` → mount path `/config` The **data** folder from step 1, not the one above it. With write access, not read-only.
Environment	three variables, see below

Under **Environment** you add three entries:

```
ENVIRONMENT VARIABLES                    PER LINE: NAME, THEN VALUE

FLOWBRIDGE_CONFIG     /config/config.yaml
TZ                    Europe/Berlin
FLOWBRIDGE_PASSWORD   your-own-password
```

DO NOT LEAVE OUT THE PASSWORD

FlowBridge can not only read your power station but also control it — start charging, switch outputs. Without this variable it runs unprotected on the network at first start, until someone sets one in the browser.

Here **without** quotation marks — unlike in the text block of route A. The wizard takes the value literally; quotation marks would become part of the password.

3

Start

Next, glance over the summary, then **Done**. Leave the tick at „Run this container after the wizard is finished“ set.

Under **Container**, `flowbridge` should show **Running** right afterwards.

DOES IT SAY „STOPPED“?

Then the **Log** tab says why, in plain words. The most common case is the folder: if the mount does not point at `data`, or write access is missing, FlowBridge cannot store its configuration.

THE SAME FOR BOTH ROUTES

Setting up FlowBridge

The container is running — it just does not know anything about your power station yet. That comes now, in the browser.

1 Open the interface for the first time

Open a new browser tab and enter the address of your NAS, followed by a colon and `8081`:

```
http://192.168.1.20:8081
```

Replace the numbers with the IP of your own NAS, of course — the same one you use to reach DSM.

A sign-in appears. Enter the password you set as `FLOWBRIDGE_PASSWORD` during setup.

DOES IT SAY „SET PASSWORD“ INSTEAD?

Then FlowBridge did not read your line — usually a typo in the name `FLOWBRIDGE_PASSWORD`. No great matter: just set the password here in the interface and carry on.

2 Enter EcoFlow and broker

You land straight in **Setup**. Five fields, top to bottom:

Setup	
Access key	from the EcoFlow developer portal
Secret key	from there as well
Local broker IP	`192.168.1.50` The address of your MQTT broker, not that of the NAS — unless the broker runs on the NAS.
Port	`1883` Usually filled in already.
User name, password	only if your broker requires them Otherwise leave both fields empty.

Fetch the device instead of typing it

As soon as both keys are in, click **Find devices**. FlowBridge queries your EcoFlow account and fills in serial number and model by itself — typing serial numbers is a source of errors you can spare yourself.

You may overwrite the name. It shows up later in the interface and in the MQTT topics, so pick something telling, like `Garage battery`.

Test first, then save

Click **Test connection**. If it works, FlowBridge reports the EcoFlow broker and how many readings it found — for a River 2 Pro that is around a hundred. **If a number is shown, everything is right.**

Then click **Save**.

3 Check that it flows

After saving, FlowBridge jumps to the dashboard. At the top you see three status indicators: **EcoFlow**, **MQTT** and your power station. All three green means: readings are arriving and going back out.

State of charge, input and output power should now move every second. On your broker the values sit under topics beginning with `flowbridge/`.

HOME ASSISTANT

If HA runs on the same broker, you need to configure nothing there. FlowBridge announces itself, and your power station appears as a device with all its sensors.

That is it. FlowBridge is running.

AFTERWARDS

Two things for later

Take the password out of the settings

It sits there in the clear. Once FlowBridge has run, you can delete the `FLOWBRIDGE_PASSWORD` line — depending on which route you took:

Route A: Project › flowbridge › Action › Edit

Route B: Container › flowbridge › Details › Environment

Your password is unaffected by this — it has long been stored encrypted in the data folder, and this line is only read at the very first start.

Updating to a new version

FlowBridge tells you itself when there is something new: at the top right it then says **Update available** with the number. Clicking it checks immediately instead of waiting for the next round.

You fetch it by **entering the number**. So not through a button, but through a line or a field.

Route A — in the project

```
Project › flowbridge › Action › Edit
```

Change the `image` line to the number FlowBridge reported — so

```
image: cheetahlab/flowbridge:latest
```

becomes, for example

```
image: cheetahlab/flowbridge:2026.08.16-46
```

Save, then build again:

```
Project › flowbridge › Action › Build
```

Route B — via the registry

First fetch the new image: **Registry**, search for `cheetahlab/flowbridge`, **Download**, and this time pick the **new number** rather than `latest`.

Then replace the old container: under **Container** stop and delete `flowbridge`, then under **Image** start the new one with **Run** — with the same settings as the first time.

> **YOUR DATA SURVIVES THIS**
>
> Deleting affects only the container, not the `data` folder. Configuration, password and remembered setpoints live there and outlast the exchange.
>
> That this route is more cumbersome lies in the nature of the thing: a container is bound to one image. That is exactly what a project takes off your hands — it remembers the settings and swaps only the image.

Your settings and credentials live in the `data` folder and are kept throughout. Whether it worked is shown at the bottom right in FlowBridge: the running version is displayed there.

WHY THE NUMBER AND NOT `LATEST`

Because with `latest` the NAS checks whether it already has an image **of that name** — and if so, takes the one it has. The container restarts, everything looks right — and you carry on running the old version. That is the unpleasant kind of fault, because it does not announce itself.

A version number, by contrast, the NAS has never seen. It **has** to fetch it. That is why this is the route that always works.

There is also **Image › Update**, and that fetches too. Only, the **Update available** hint next to the image appears just once the NAS has looked of its own accord — which it does on its own schedule, not when you open the page. Shortly after a new version there is often nothing there yet, although there has long been something. Entering the number waits for nobody.

DO NOT USE „RESET“

The obvious choice would be **Action › Reset**. That fetches **nothing** — it only restarts, with the same image as before.

GOING BACK TO AN OLDER VERSION

Should a new version break something: the same line, only with the old number.

```
image: cheetahlab/flowbridge:2026.08.16-39
```

Version numbers are set **immutable** on Docker Hub — under a number once given lies exactly the same image forever. A rollback therefore takes you reliably to the state that ran before, not to something that merely goes by the same name.

Do not switch back to `latest` while doing so. It sounds obvious, but it throws you back silently: under that name the NAS still holds whatever version you last fetched through it — possibly months old. Stay with the number and enter the new one next time. You get the update hints regardless; those come from FlowBridge itself.

ROUTE C – FOR THE CURIOUS

Building from source yourself

FlowBridge is free software, the complete source is open. Whoever wants to look inside or change something builds the image themselves — then nothing runs at your place that you did not compile yourself.

This is **not a third installation route for the NAS**: building happens on a machine with Docker and Git, not in Container Manager. For normal operation it gains you nothing — the finished image is the same.

ON YOUR OWN MACHINE — DOCKER AND GIT REQUIRED

```
# fetch the source
git clone https://github.com/CheetahLab/FlowBridge.git
cd FlowBridge

# enable the version hook – otherwise the interface
# will later report an outdated build
git config core.hooksPath scripts/githooks

# build and start
docker compose -f docker/flowbridge/compose.build.yaml up -d --build
```

After that the procedure is the same as above: open the interface in the browser, enter the EcoFlow keys and the broker.

WHERE EVERYTHING ELSE IS DESCRIBED

The repository `CheetahLab/FlowBridge` on GitHub carries a detailed description: which file does what, how the tests run, what the EcoFlow interface provides — and what it does not.

WHEN IT STICKS

Troubleshooting

Click on whatever matches your case.

- **The interface does not come up at all**

First check in Container Manager whether the container really shows **Running**. If it keeps restarting, the **Log** tab helps — the cause is stated there in plain words.

If it runs but the browser does not get through, check the IP address. Perhaps port `8081` is already taken by another service. Then change the **left** number, to `8091` say — on route A in the project file (`"8091:8080"`), on route B in the port settings of the container. Afterwards call the address with `:8091` .

„Test connection“ fails

Almost always one of three causes:

- A space slipped in when copying the keys — at the front or the back. Best to paste them again.
- The power station belongs to a **different EcoFlow account** than the one you generated the keys for.
- The device is currently offline and unreachable for the cloud. Check in the EcoFlow app whether it appears there.

Readings arrive but do not reach the broker

Then the EcoFlow side is up and the local one is not. Check the IP and port of the broker and whether it demands credentials.

Typical case: Mosquitto with `allow_anonymous false`. Then you need a user name and password in FlowBridge as well.

Password forgotten

No drama, that can be reset. In File Station open this file:

```
docker › flowbridge › data › config.yaml
```

Delete the `auth:` section in it — together with all the indented lines directly below it. Save, restart the container. On the next visit FlowBridge asks for a new password again; everything else is preserved.

- **None of this helps — send a diagnostics package**

For that there is the **Diagnostics** card in **Settings**. The order matters:

1. Switch logging on
2. Trigger the fault once more
3. Download the diagnostics package and send it by e-mail

It contains the version, the state of all connections and the log — everything needed for remote analysis.

Your keys and passwords are redacted in it, and indeed before anything is written at all. You can send the file without hesitation.

FlowBridge — MQTT bridge for portable power stations by ECOFLOW.

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Independent project, not affiliated with EcoFlow. FlowBridge speaks unencrypted HTTP — on your own network that is fine. Whoever wants to make it reachable from the internet puts a reverse proxy with TLS in front.